ENIVIRONMENTAL CHEMISTRY

Earth's System

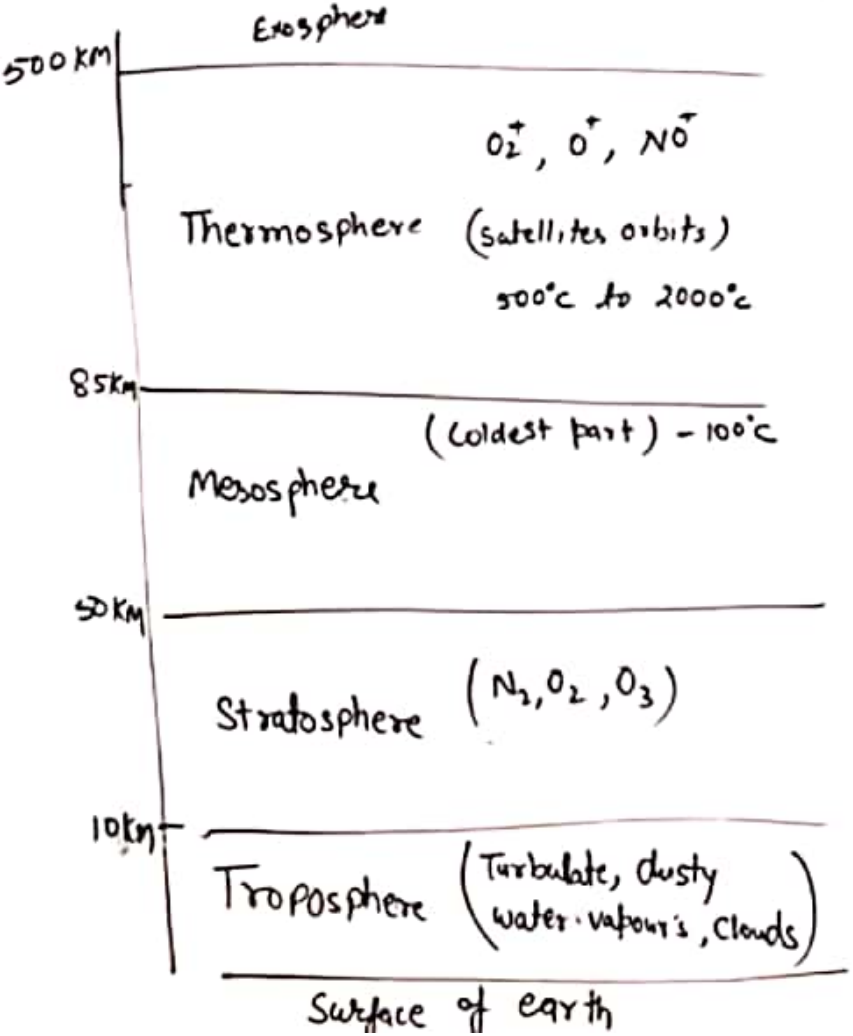
- **Atmosphere** (Layers of air that surrounds the earth)
- **Hydrosphere** (Water portion on earth)
- **Lithosphere** (Except hydrosphere rest of the earth space)
- **Biosphere** All living organism (Plants, animals, micro organism)

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Atmosphere:- It is a protective blanket of gases surrounding the earth



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Pollutants:- A substance, which causes pollution.

- Solid pollutant
- Liquid pollutant
- gas pollutant

- Degradable pollutant (which rapidly break down by natural processes.)
- Non-degradable = (Remain in the environment in an unchanged form for many decades.
  eg DDT, Plastic material, nuclear wastes etc.

⇒ Tropospheric pollutant
- Particulated (Solid & liq. droplets in air) eg dust, Soot, Mist, fume etc.
- Gaseous pollutant (oxides of S, N, C etc)

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Gaseous air Pollutants:-

⇒ Oxides of Sulpher

$SO_2 + O_2 \longrightarrow SO_3$

$SO_2 + O_3 \longrightarrow SO_3 + O_2$

$SO_2 + H_2O_2 \longrightarrow H_2SO_4$

Toxic effect:-

Headache, vomiting, irritation of eyes, respiratory desease

→ Source of acid rain

⇒ Oxides of nitrogen:-

$N_2 + O_2 \xrightarrow[\text{lightning strikes}]{1483\,K} 2NO$

$NO + O_2 \xrightarrow{(Slow)} NO_2$

$NO + O_3 \xrightarrow{(Fast)} NO_2 + O_2$

→ Damages the leaves of plants

→ Slow rate of photosynthesis

→ lung irritant

→ Toxic to living tissues.

⇒ Oxides of carbon:-

- CO : Bind to haemoglobin and reduces the oxygen carrying capacity of blood to greater extent.
  → form photochemical smog.
- CO_2
  mainly responsible for global warming.

⇒ Hydrocarbons

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Global warming and Greenhouse effect:-

Gasese such a CO_2, CH_4, O_3, N_2O (nitrous oxide), water vapours, CFCs allow sun rays to pass through them but then absorb and reradiate the heat back toword the earth. These are therefore termed as green house gases.

- Global warming is difined as the increase in the average global temperature of the atmosphere near earth's surface due to presence of green house gases.

Harmful effects:

1. Melting of polar ice cups.
2. Increase global temp and Climate change.
3. Effect on growth of plants and animals.
4. Increase infectious diseases.

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ACID RAIN

When the pH of rain water drops below 5.6, it is called acid rain.

Nowadays, the term acid rain is used to describe the way in which acid from the atmosphere is deposited on the earth's surface.

$CO_2 + H_2O \longrightarrow H_2CO_3$ (Carbonic acid)

It mainly consists of oxides of S and N.

Source of pollutants:-

1. Burning of fossil fuels, such a coal and oil in power station and furnaces or petrol, diesel in motor engines.
2. Volcanic eruption, forest fires and bacterial decomposition of organic matter.

Reactions

$$SO_2 + O_2 + H_2O \longrightarrow H_2SO_4$$

$$NO_2 + O_2 + H_2O \longrightarrow HNO_3$$

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ACID RAIN

Harmful effect of acid rain:-

① Acid rain is harmful for agriculture, plants and trees.

② Acid rain affects plants and animal life in aquatic ecosystem.

③ It cause respiratory ailments in human beings and animals.

④ It corrodes water pipes resulting in the dissolution of heavy metals such as Fe, Pb, & Cu into the drinking water.

⑤ Building materials like, stone, marble, dolomite, metal are corroded and weakened on reaction with acid rain because of the formation of soluble salt.

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Smog

- Smog word is combination of smoke and fog.

① Classical smog :- It occurs in cool and humid climate

- It is a mixture of smoke, fog & SO_2 (Sulphurous smog)
- due to presence of SO_2 and carbon partical it is a reducing mixture (Reducing smog)

② Photochemical smog :-

- Occurs in warm, dry & sunny climate.
- Components = unsaturated hydrocarbons, nitrogen oxide.
- due to presence of NO_2 it is oxidising in nature (oxidising smog.)

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SMOG

Formation of Photochemical smog :-

Automobile exhousts give out NO_2 and hydrocarbon. They form PAN (Peroxy-acetyl nitrate) which along with ozone forms photochemical smog

$$NO_2 \xrightarrow{h\nu} NO + O$$

$$O_2 + O \rightleftharpoons O_3$$

$$NO + O_3 \longrightarrow NO_2 + O_2$$

NO_2 and O_3 are toxic gases and strong oxidising agents. they can react with unburnt hydrocarbons in the polluted air to produce compounds of photochemical smog

$3CH_4 + 2O_3 \longrightarrow 3CH_2=O + 2O_2$; $CH_2=CH-CHO$ (acrolein)

$CH_3-\overset{\overset{O}{\|}}{C}-O-ONO_2$ (Peroxyacetyl nitrate) PAN

⇒ Harmful effect of photochemical smog :-

- Respiratory problems.
- Headache
- Chest Pain
- irritation in eyes
- Smog damage crops
- Cracking of rubber

⇒ Primary precursors = NO_2 and hydrocarbon

⇒ Secondary Precursors = O_3 and PAN

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STRATOSPHERIC POLLUTION

Ozone depletion:-

The upper part of Stratosphere consists of considerable amount of ozone. this layer protects us from the harmful ultraviolate radiation ($\lambda = 255\,nm$)

→ Ozone is produced as result of the action of Ultraviolate radiation of O_2

$$O_2 \xrightarrow{UV} 2O$$

$$O + O_2 \overset{UV}{\rightleftharpoons} O_3$$

→ The main reason of ozone layer depletion is release of chlorofluorocarbons (CFCs : Freon)

$$CF_2Cl_2 \xrightarrow{UV} \dot{Cl} + \dot{C}F_2Cl$$

$$\dot{Cl} + O_3 \rightarrow \dot{C}lO + O_2$$

$$\dot{C}lO + O \rightarrow \dot{Cl} + O_2$$

Harmful effects of ozone hole:-

- Plants and animals living on the surface will start dying
- UV radiation will fastern the formation of smog.
- Temperature of the earth will increase.
- Skin cancer

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WATER POLLUTION :-

⇒ Contamination of Streams, lakes, sea, underground water.

⇒ Physical parameters for polluted water :- Colour, Odour, turbidity etc.

⇒ Chemical parameters " " " :- amount of carbonate, sulphate, Chloride, $F^{\ominus}$, nitrate and metal ions.

Lead	0 – 50 PPb
$F^{\ominus}$	1 – 2 PPm
NO_3^-	0 – 50 PPm
SO_4^{-2}	0 – 500 PPM

maximum prescribed concentration of metal in drinking water

Cd = 0.005 PPm
Mn = 0.05 PPm
Al = 0.2 PPm
Fe = 0.2 PPM
Cu = 3 PPm
Zn = 5 PPm

* $[3(Ca_3(PO_4)_2 . Ca(OH)_2] \xrightarrow{F^-} [3\ Ca_3(PO_4)_2 CaF_2]$

Hydroxyapatite → Fluorapatite

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WATER POLLUTION

⇒ Causes of water pollution :
- 1. Chemical Pollution
- 2. Pathogens
- 3. Organic Wastes

2. Pathogens :-

- Bacteria, virus or other microognism that can cause disease are called 'Pathogens'
- ⇒ Most serious water pollutants
- ⇒ Sourse :- Domestic sewage and animal excreta.
- ⇒ Human excreta contain bacteria such as E. Coli and Streptococcus faecolis which cause gastrointestinal disease.

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WATER POLLUTION

(3) Organic wastes.

⇒ Biodegradable waste.

⇒ Due to leaves, grass, excessive aquatic plant growth.

Dissolved oxygen (D.O.):
- From atmosphearic air.
- Photolysis of aquatic plants.

- D.O. is important for aquatic life.
- Max. D.O. for cold water 10 ppm (oxygen in air 200000 ppm)
- The large population of bacteria decomposes organic metter present in water, they consume oxygen of water
- If D.O. below 6 ppm the growth of fish inhibited

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WATER POLLUTION

(3) Organic wastes.

High organic waste → high Polulation of microorganism → D.O ↓es becomes zero

- becomes zero → Anaerobic bacteria active → Decomposition of organic wastes. → Produce Chemicals (foul smell) → Harmful for our health.
- becomes zero → aquatic life end

Biochemical oxygen demand (BOD):- The amount of oxygen required by bacteria to break down the organic matter present in a certain volume of water sample.

⇒ for clean water BOD less than 5 PPM → if BOD 17 or more = highly polluted water.

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SOIL POLLUTION :-

⇒ India is agriculture based economy.

⇒ The surplus production of products of crops are stored by governmental and non governmental organisations for the lean seasons.

⇒ We use many insecticides, Pesticides and herbicides to protect crops and products of crops for storage.

- These Perticides, insecticides and herbicides are cause of soil pollution.

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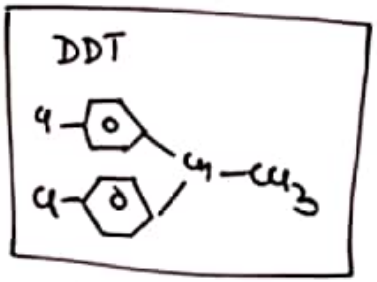
SOIL POLLUTION :-

⇒ Pesticides :-

- Pesticides are synthetic toxic chemicals with ecological repercussion.

  Example :- DDT, Aldrin, Dieldrin, Malathion, Diazinon, glyphosate etc.

- Repeated use of same of same pesticide give rise to pests that are resistant to that group of pesticides thus making the pesticides ineffective.



- Most of pesticides are water insoluble and non biodegradable.
- Therefore increase concentration in higher trophic level through the food chain.
- Organo-Phosphates and carbamates are less persistant, more biodegradable, but more harmful.

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SOIL POLLUTION :-

Herbicides:-

A herbicide is used to Kill unwanted plants.

selective herbicide Kill certain targets while leaving the desired crop relatively unharmed.

eg $NaClO_2$ Na_3AsO_3 etc.

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Industrial Waste

ⓐ Biodegradable :-

Generated by

- Cotton mills
- Food processing units
- Paper mills
- Textile factories.

ⓑ Non-biodegradable :-

Generated by

- Thermal power plants (fly ash)
- Integrated iron, steel plants (Blast furnanse slag)
- Al, Zn, Cu, Manufacturing plants (mud, tailing)
- Fertilizer industry (gypsum)
- Chemical industry
- Drug, Pharmaceuticals industries.
- dyes factories
- rubber factories.

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